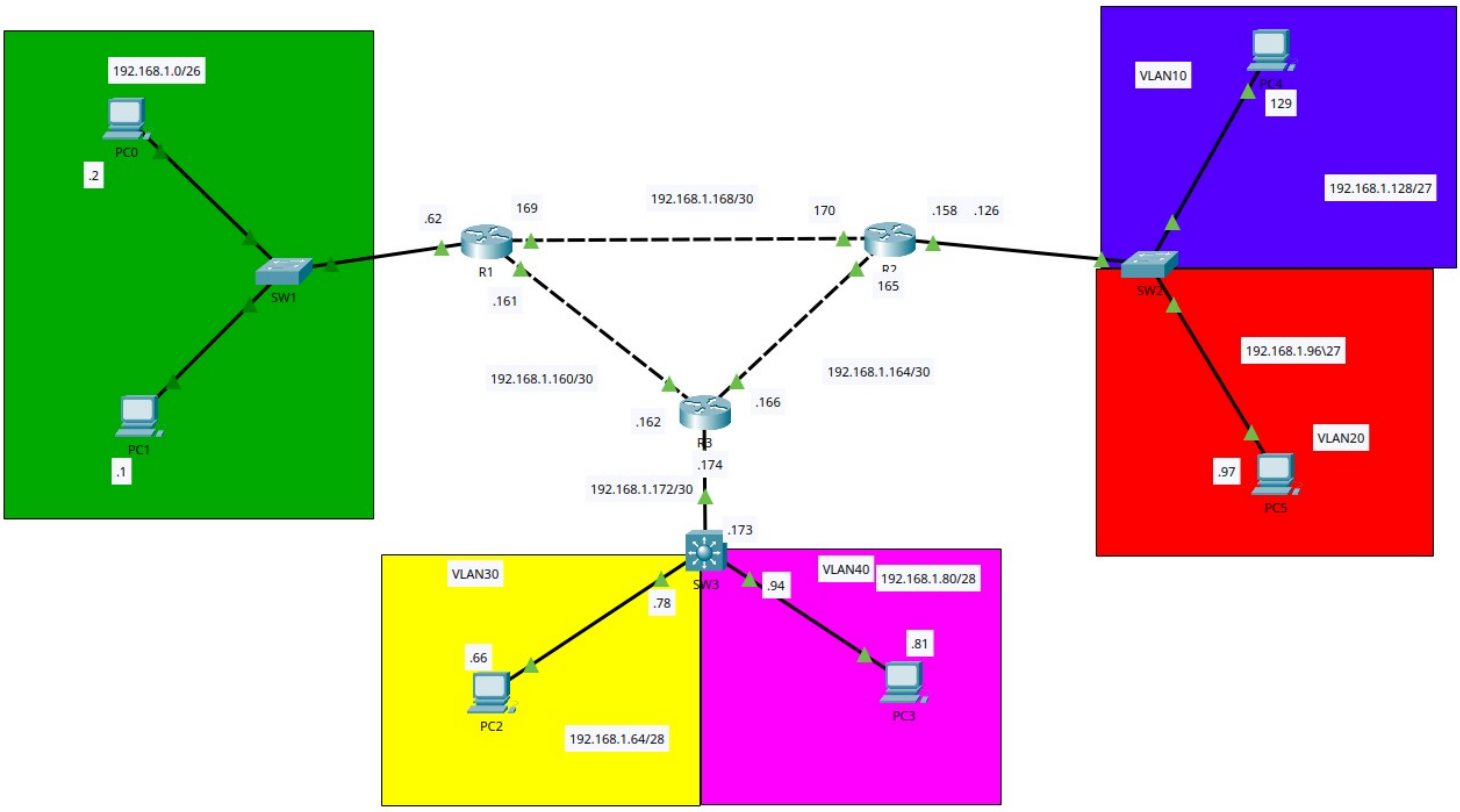
# Scheme




The network topology is basic. It features 3 routers. After all of them is a Switch. The left switch has no configurations. The other 2 switches use VLANs. The right switch uses simple trunk port to the router. The switch in the middle is multilayer switch.

# Router Configuration

This is the configuration for the right router. It features the sub-interface method to connect to the VLANs via ROAS. The other ones follow the same pattern, except they don't have sub-interfaces configured, only physical.

```
Router>en\
Translating "en\"...domain server (255.255.255.255) % Name lookup aborted
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#int g0/0
R2(config-if)#int g0/0.10
R2(config-subif)#enc
R2(config-subif)#encapsulation dot
R2(config-subif)#encapsulation dot1Q 10
R2(config-subif)#ip add 192.168.1.158 255.255.255.224
R2(config-subif)#no sh
R2(config-subif)#int g0/0.20
R2(config-subif)#ip add 192.168.1.126 255.255.255.224

% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL vLAN.

R2(config-subif)#encapsulation d
R2(config-subif)#encapsulation dot1Q 20
R2(config-subif)#ip add 192.168.1.126 255.255.255.224
R2(config-subif)#no sh
R2(config-subif)#
R2(config-subif)#int g0/2
R2(config-if)#ip add 192.168.1.165 255.255.255.252
R2(config-if)#no sh

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

R2(config-if)#int g0/1
R2(config-if)#ip add 192.168.1.170 255.255.255.252
R2(config-if)#no sh

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
```

# Trunk port configuration on right switch

```
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                               Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                               Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                               Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                               Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                               Fa0/23, Fa0/24, Gig0/2
10   VLAN0010                         active    Fa0/1
20   VLAN0020                         active    Fa0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

```
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#int g0/0
%Invalid interface type and number
SW2(config)#swi
SW2(config)#swit
SW2(config)#int g0/0
%Invalid interface type and number
SW2(config)#int g0/1
SW2(config-if)#swit
SW2(config-if)#switchport mo
SW2(config-if)#switchport mode tr
SW2(config-if)#switchport mode trunk
SW2(config-if)#switchport tru
SW2(config-if)#switchport trunk al
SW2(config-if)#switchport trunk allowed vl
SW2(config-if)#switchport trunk allowed vlan 10,20
SW2(config-if)#do sh int trunk
```

This router uses simple trunk. It has the 2 VLANs connected directly to it through access ports.

# Middle switch

The middle switch is a multilayer switch. I configured it to use Switch Virtual Interfaces.

```
GigabitEthernet0/2     unassigned      YES unset  down                    down
Vlan1                  unassigned      YES unset  administratively down down
Vlan30                 192.168.1.78    YES manual up                      up
Vlan40                 192.168.1.94    YES manual up                      up
SW3#
SW3#
```

```
30   VLAN0030                         active    Fa0/1
40   VLAN0040                         active    Fa0/2
```

For the connection between this switch and the router I used default route on the switch to the router, and static routes on the router

```
SW3#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.1.174 to network 0.0.0.0

     192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
C       192.168.1.64/28 is directly connected, FastEthernet0/1
C       192.168.1.80/28 is directly connected, FastEthernet0/2
C       192.168.1.172/30 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 192.168.1.174

SW3#
```

```
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.1.0/24 is variably subnetted, 12 subnets, 5 masks
O       192.168.1.0/26 [110/2] via 192.168.1.161, 00:10:43, GigabitEthernet0/1
S       192.168.1.64/28 [1/0] via 192.168.1.173
S       192.168.1.80/28 [1/0] via 192.168.1.173
```

# Routing - RIP

I did the routing between LANs with RIP and OSPF. This is a routing table using RIP.

```
R2>en
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.1.0/24 is variably subnetted, 13 subnets, 5 masks
R       192.168.1.0/26 [120/1] via 192.168.1.169, 00:00:04, GigabitEthernet0/1
R       192.168.1.64/28 [120/2] via 192.168.1.166, 00:00:06, GigabitEthernet0/2
R       192.168.1.80/28 [120/2] via 192.168.1.166, 00:00:06, GigabitEthernet0/2
C       192.168.1.96/27 is directly connected, GigabitEthernet0/0.20
L       192.168.1.126/32 is directly connected, GigabitEthernet0/0.20
C       192.168.1.128/27 is directly connected, GigabitEthernet0/0.10
L       192.168.1.158/32 is directly connected, GigabitEthernet0/0.10
R       192.168.1.160/30 [120/1] via 192.168.1.166, 00:00:06, GigabitEthernet0/2
                         [120/1] via 192.168.1.169, 00:00:04, GigabitEthernet0/1
C       192.168.1.164/30 is directly connected, GigabitEthernet0/2
L       192.168.1.165/32 is directly connected, GigabitEthernet0/2
C       192.168.1.168/30 is directly connected, GigabitEthernet0/1
L       192.168.1.170/32 is directly connected, GigabitEthernet0/1
R       192.168.1.172/30 [120/1] via 192.168.1.166, 00:00:06, GigabitEthernet0/2

R2#
```

# Routing - OSPF

After configuring with RIP I removed it and configured OSPF on the same topology. It has only 1 area. This is the routing table of R3 with OSPF.

```
     192.168.1.0/24 is variably subnetted, 12 subnets, 5 masks
O       192.168.1.0/26 [110/2] via 192.168.1.161, 00:38:05, GigabitEthernet0/1
S       192.168.1.64/28 [1/0] via 192.168.1.173
S       192.168.1.80/28 [1/0] via 192.168.1.173
O       192.168.1.96/27 [110/2] via 192.168.1.165, 00:35:55, GigabitEthernet0/2
O       192.168.1.128/27 [110/2] via 192.168.1.165, 00:36:05, GigabitEthernet0/2
C       192.168.1.160/30 is directly connected, GigabitEthernet0/1
L       192.168.1.162/32 is directly connected, GigabitEthernet0/1
C       192.168.1.164/30 is directly connected, GigabitEthernet0/2
L       192.168.1.166/32 is directly connected, GigabitEthernet0/2
O       192.168.1.168/30 [110/2] via 192.168.1.161, 00:37:00, GigabitEthernet0/1
                         [110/2] via 192.168.1.165, 00:37:00, GigabitEthernet0/2
C       192.168.1.172/30 is directly connected, GigabitEthernet0/0
L       192.168.1.174/32 is directly connected, GigabitEthernet0/0
```

Also configured the router IDs. This are the neighbours of R3.

```
R3#sh ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/BDR        00:00:34    192.168.1.161   GigabitEthernet0/1
2.2.2.2           1   FULL/BDR        00:00:31    192.168.1.165   GigabitEthernet0/2
```

And since the R3 has static routes to networks that are not directly connected I had to use the command ‘redistribute static subnets’ to get the router to advertise them.